

BLM2041 Signals and Systems

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Digital Signal Processing

FIR Filtering and Frequency Response

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LECTURE OBJECTIVES

- INTRODUCE FILTERING IDEA
 - Weighted Average
 - Running Average
- FINITE IMPULSE RESPONSE FILTERS
 - **FIR** Filters
 - Show how to **compute** the output $y[n]$ from the input signal, $x[n]$

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LECTURE OBJECTIVES

- **SINUSOIDAL** INPUT SIGNAL
 - DETERMINE the **FIR FILTER** OUTPUT
- **FREQUENCY RESPONSE** of FIR
 - PLOTTING vs. Frequency
 - MAGNITUDE vs. Freq
 - PHASE vs. Freq

$$H(e^{j\omega}) = |H(e^{j\omega})| e^{j\angle H(e^{j\omega})}$$

MAG

PHASE

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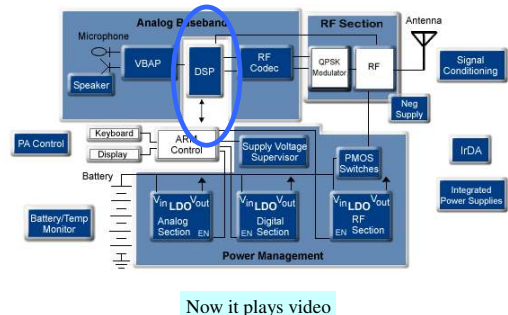
DIGITAL FILTERING



- CONCENTRATE on the COMPUTER
 - PROCESSING ALGORITHMS
 - SOFTWARE (MATLAB)
 - HARDWARE: DSP chips, VLSI
- **DSP**: DIGITAL SIGNAL PROCESSING

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Digital Cell Phone



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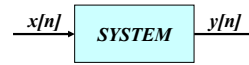
DISCRETE-TIME SYSTEM



- OPERATE on $x[n]$ to get $y[n]$
- WANT a **GENERAL** CLASS of SYSTEMS
 - **ANALYZE** the **SYSTEM**
 - **TOOLS**: TIME-DOMAIN & FREQUENCY-DOMAIN
 - **SYNTHESIZE** the **SYSTEM**

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D-T SYSTEM EXAMPLES

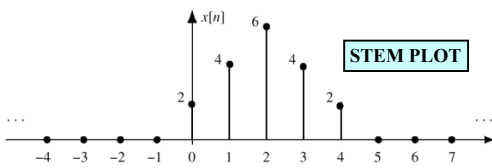


- **EXAMPLES**:
 - **POINTWISE OPERATORS**
 - **SQUARING**: $y[n] = (x[n])^2$
 - **RUNNING AVERAGE**
 - **RULE**: “the output at time n is the average of three consecutive input values”

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DISCRETE-TIME SIGNAL

- $x[n]$ is a **LIST** of **NUMBERS**
 - **INDEXED** by “ n ”



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3-PT AVERAGE SYSTEM

- **ADD 3 CONSECUTIVE NUMBERS**
 - **Do this for each “ n ”**

the following input–output equation

Make a TABLE

$$y[n] = \frac{1}{3}(x[n] + x[n+1] + x[n+2])$$

n	$n < -2$	-2	-1	0	1	2	3	4	5	$n > 5$
$x[n]$	0	0	0	2	4	6	4	2	0	0
$y[n]$	0	$\frac{2}{3}$	2	4	$\frac{14}{3}$	4	2	$\frac{2}{3}$	0	0

$$n=0 \quad y[0] = \frac{1}{3}(x[0] + x[1] + x[2])$$

$$n=1 \quad y[1] = \frac{1}{3}(x[1] + x[2] + x[3])$$

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3-PT AVERAGE SYSTEM

INPUT SIGNAL

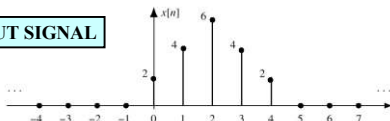


Figure 5.2 Finite-length input signal, $x[n]$.

$$y[n] = \frac{1}{3}(x[n] + x[n+1] + x[n+2])$$

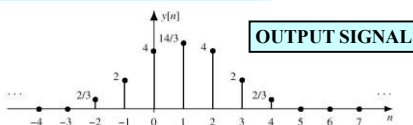


Figure 5.3 Output of running average, $y[n]$.

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PAST, PRESENT, FUTURE

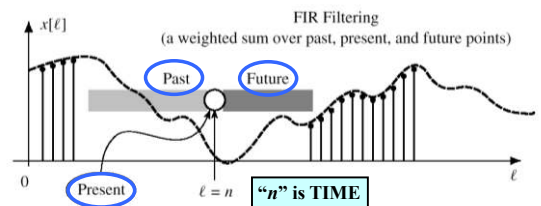


Figure 5.4 The running-average filter calculation at time index n uses values within a sliding window (shaded). Dark shading indicates the future ($\ell > n$); light shading, the past ($\ell < n$).

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ANOTHER 3-pt AVERAGER

- Uses “PAST” VALUES of $x[n]$
 - IMPORTANT IF “ n ” represents REAL TIME
 - WHEN $x[n]$ & $y[n]$ ARE STREAMS

$$y[n] = \frac{1}{3}(x[n] + x[n-1] + x[n-2])$$

n	$n < -2$	-2	-1	0	1	2	3	4	5	6	7	$n > 7$
$x[n]$	0	0	0	2	4	6	4	2	0	0	0	0
$y[n]$	0	0	0	$\frac{2}{3}$	2	4	$\frac{14}{3}$	4	2	$\frac{2}{3}$	0	0

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GENERAL FIR FILTER

- FILTER COEFFICIENTS $\{b_k\}$
 - DEFINE THE FILTER

$$y[n] = \sum_{k=0}^M b_k x[n-k]$$

– For example, $b_k = \{3, -1, 2, 1\}$

$$y[n] = \sum_{k=0}^3 b_k x[n-k] = 3x[n] - x[n-1] + 2x[n-2] + x[n-3]$$

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GENERAL FIR FILTER

- FILTER COEFFICIENTS $\{b_k\}$

$$y[n] = \sum_{k=0}^M b_k x[n-k]$$

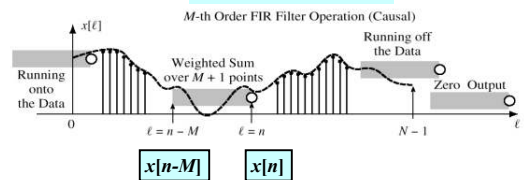
- FILTER ORDER is M
- FILTER LENGTH is $L = M+1$
 - NUMBER of FILTER COEFFS is L

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GENERAL FIR FILTER

- SLIDE a WINDOW across $x[n]$

$$y[n] = \sum_{k=0}^M b_k x[n-k]$$



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FILTERED STOCK SIGNAL



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SPECIAL INPUT SIGNALS

- $x[n] = \text{SINUSOID}$ **FREQUENCY RESPONSE (LATER)**
- $x[n]$ has only one NON-ZERO VALUE

$$\delta[n] = \begin{cases} 1 & n = 0 \\ 0 & n \neq 0 \end{cases}$$

UNIT-IMPULSE



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UNIT IMPULSE SIGNAL $\delta[n]$

n	...	-2	-1	0	1	2	3	4	5	6	...
$\delta[n]$	0	0	0	1	0	0	0	0	0	0	0
$\delta[n-3]$	0	0	0	0	0	0	1	0	0	0	0

$\delta[n]$ is NON-ZERO
When its argument
is equal to ZERO

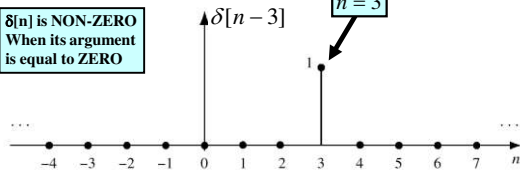


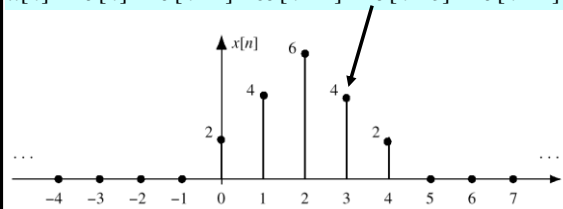
Figure 5.7 Shifted impulse sequence, $\delta[n-3]$.

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MATH FORMULA for $x[n]$

- Use **SHIFTED** IMPULSES to write $x[n]$

$$x[n] = 2\delta[n] + 4\delta[n-1] + 6\delta[n-2] + 4\delta[n-3] + 2\delta[n-4]$$



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SUM of **SHIFTED** IMPULSES

n	...	-2	-1	0	1	2	3	4	5	6	...
$2\delta[n]$	0	0	0	2	0	0	0	0	0	0	0
$4\delta[n-1]$	0	0	0	0	4	0	0	0	0	0	0
$6\delta[n-2]$	0	0	0	0	0	6	0	0	0	0	0
$4\delta[n-3]$	0	0	0	0	0	0	4	0	0	0	0
$2\delta[n-4]$	0	0	0	0	0	0	0	2	0	0	0
$x[n]$	0	0	0	2	4	6	4	2	0	0	0

$$x[n] = \sum_k x[k]\delta[n-k]$$

← This formula ALWAYS works

$$= \dots + x[-1]\delta[n+1] + x[0]\delta[n] + x[1]\delta[n-1] + \dots \quad (5.3.6)$$

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4-pt AVERAGER

- CAUSAL SYSTEM: USE PAST VALUES

$$y[n] = \frac{1}{4}(x[n] + x[n-1] + x[n-2] + x[n-3])$$

- INPUT = UNIT IMPULSE SIGNAL = $\delta[n]$

$$x[n] = \delta[n]$$

$$y[n] = \frac{1}{4}\delta[n] + \frac{1}{4}\delta[n-1] + \frac{1}{4}\delta[n-2] + \frac{1}{4}\delta[n-3]$$

- OUTPUT is called **"IMPULSE RESPONSE"**

$$h[n] = \{\dots, 0, 0, \frac{1}{4}, \frac{1}{4}, \frac{1}{4}, \frac{1}{4}, 0, 0, \dots\}$$

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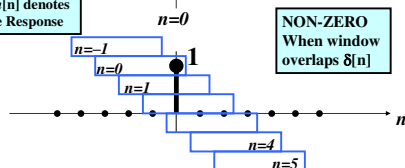
4-pt Avg Impulse Response

$$y[n] = \frac{1}{4}(x[n] + x[n-1] + x[n-2] + x[n-3])$$

$\delta[n]$ "READS OUT" the FILTER COEFFICIENTS

$$h[n] = \{\dots, 0, 0, \frac{1}{4}, \frac{1}{4}, \frac{1}{4}, \frac{1}{4}, 0, 0, \dots\}$$

"h" in $h[n]$ denotes
Impulse Response



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FIR IMPULSE RESPONSE

- Convolution = Filter Definition

– Filter Coeffs = Impulse Response

n	$n < 0$	0	1	2	3	...	M	$M+1$	$n > M+1$
$x[n] = \delta[n]$	0	1	0	0	0	0	0	0	0
$y[n] = h[n]$	0	b_0	b_1	b_2	b_3	...	b_M	0	0

$$y[n] = \sum_{k=0}^M b_k x[n-k]$$

$$y[n] = \sum_{k=0}^M h[k] x[n-k]$$

CONVOLUTION

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FILTERING EXAMPLE

- 7-point AVERAGER

– Removes cosine

- By making its amplitude (A) smaller

$$y_7[n] = \sum_{k=0}^6 \left(\frac{1}{7}\right)x[n-k]$$

- 3-point AVERAGER

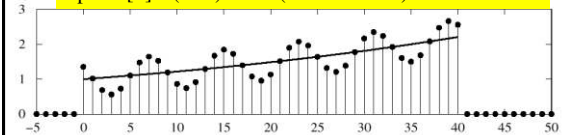
– Changes A slightly

$$y_3[n] = \sum_{k=0}^2 \left(\frac{1}{3}\right)x[n-k]$$

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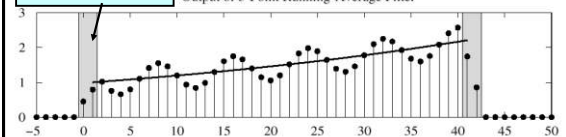
3-pt AVG EXAMPLE

Input: $x[n] = (1.02)^n + \cos(2\pi n/8 + \pi/4)$ for $0 \leq n \leq 40$



USE PAST VALUES

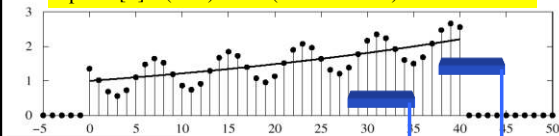
Output of 3-Point Running-Average Filter



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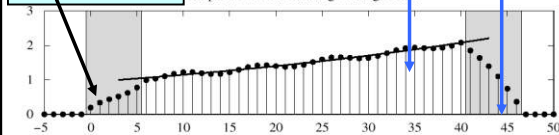
7-pt FIR EXAMPLE (AVG)

Input: $x[n] = (1.02)^n + \cos(2\pi n/8 + \pi/4)$ for $0 \leq n \leq 40$



CAUSAL: Use Previous

Output of 7-Point Running-Average Filter



LONGER OUTPUT

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Linearity & Time-Invariance, Convolution

- IMPULSE RESPONSE, $h[n]$
 - FIR case: same as $\{b_k\}$
- CONVOLUTION
 - GENERAL: $y[n] = h[n] * x[n]$
 - GENERAL CLASS of SYSTEMS
 - LINEAR and TIME-INVARIANT
- ALL LTI systems have $h[n]$ & use convolution

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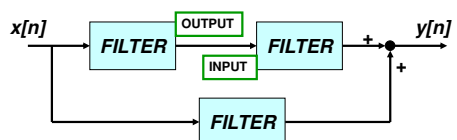
DIGITAL FILTERING



- CONCENTRATE on the FILTER (DSP)
- DISCRETE-TIME SIGNALS
 - FUNCTIONS of n , the “time index”
 - INPUT $x[n]$
 - OUTPUT $y[n]$

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BUILDING BLOCKS



- BUILD UP COMPLICATED FILTERS
 - FROM SIMPLE MODULES
 - Ex: FILTER MODULE MIGHT BE 3-pt FIR

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GENERAL FIR FILTER

- FILTER COEFFICIENTS $\{b_k\}$

– DEFINE THE FILTER

$$y[n] = \sum_{k=0}^M b_k x[n-k]$$

– For example,

$$b_k = \{3, -1, 2, 1\}$$

$$y[n] = \sum_{k=0}^3 b_k x[n-k] = 3x[n] - x[n-1] + 2x[n-2] + x[n-3]$$

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MATLAB for FIR FILTER

$$\mathbf{yy} = \text{conv}(\mathbf{bb}, \mathbf{xx})$$

– VECTOR **bb** contains Filter Coefficients

• <https://www.mathworks.com/help/matlab/ref/conv.html>

- FILTER COEFFICIENTS $\{b_k\}$

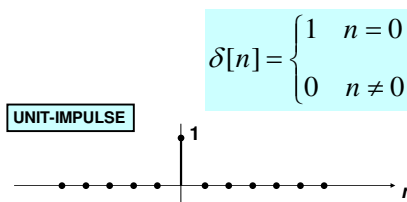
$$y[n] = \sum_{k=0}^M b_k x[n-k]$$

conv2()
for images

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SPECIAL INPUT SIGNALS

- $x[n] = \text{SINUSOID}$ **FREQUENCY RESPONSE**
- $x[n]$ has only one NON-ZERO VALUE



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FIR IMPULSE RESPONSE

- Convolution = Filter Definition
- Filter Coeffs = Impulse Response

n	$n < 0$	0	1	2	3	...	M	$M+1$	$n > M+1$
$x[n] = \delta[n]$	0	1	0	0	0	0	0	0	0
$y[n] = h[n]$	0	b_0	b_1	b_2	b_3	...	b_M	0	0

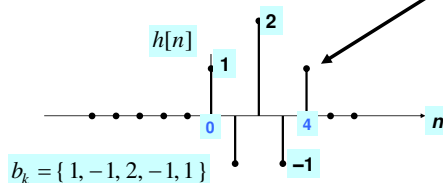
$$h[n] = \sum_{k=0}^M b_k \delta[n-k]$$

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MATH FORMULA for $h[n]$

- Use **SHIFTED** IMPULSES to write $h[n]$

$$h[n] = \delta[n] - \delta[n-1] + 2\delta[n-2] - \delta[n-3] + \delta[n-4]$$



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LTI: Convolution Sum

- Output = Convolution of $x[n]$ & $h[n]$**

– NOTATION: $y[n] = h[n] * x[n]$

– Here is the FIR case:

$$y[n] = \sum_{k=0}^M h[k] x[n-k]$$

Same as b_k

FINITE LIMITS

FINITE LIMITS

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CONVOLUTION Example

$$h[n] = \delta[n] - \delta[n-1] + 2\delta[n-2] - \delta[n-3] + \delta[n-4]$$

$$x[n] = u[n]$$

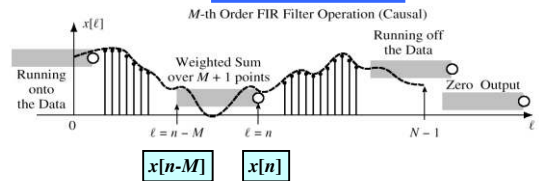
n	-1	0	1	2	3	4	5	6	7
$x[n]$	0	1	1	1	1	1	1	1	...
$h[n]$	0	1	-1	2	-1	1	0	0	0
$h[0]x[n]$	0	1	1	1	1	1	1	1	1
$h[1]x[n-1]$	0	0	-1	-1	-1	-1	-1	-1	-1
$h[2]x[n-2]$	0	0	0	2	2	2	2	2	2
$h[3]x[n-3]$	0	0	0	0	-1	-1	-1	-1	-1
$h[4]x[n-4]$	0	0	0	0	0	1	1	1	1
$y[n]$	0	1	0	2	1	2	2	2	...

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GENERAL FIR FILTER

- SLIDE a Length-L WINDOW over $x[n]$

$$y[n] = \sum_{k=0}^M b_k x[n-k]$$



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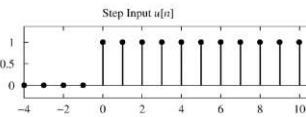
POP QUIZ

- FIR Filter is "FIRST DIFFERENCE"

$$y[n] = x[n] - x[n-1]$$

- INPUT is "UNIT STEP"

$$u[n] = \begin{cases} 1 & n \geq 0 \\ 0 & n < 0 \end{cases}$$



- Find $y[n]$

$$y[n] = u[n] - u[n-1] = \delta[n]$$

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HARDWARE STRUCTURES

$$x[n] \rightarrow \text{FILTER} \rightarrow y[n]$$

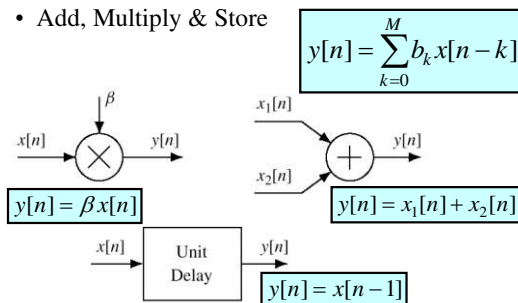
$$y[n] = \sum_{k=0}^M b_k x[n-k]$$

- INTERNAL STRUCTURE of "FILTER"
 - WHAT COMPONENTS ARE NEEDED?
 - HOW DO WE "HOOK" THEM TOGETHER?
- SIGNAL FLOW GRAPH NOTATION

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HARDWARE ATOMS

- Add, Multiply & Store



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FIR STRUCTURE

- Direct Form

$$y[n] = \sum_{k=0}^M b_k x[n-k]$$

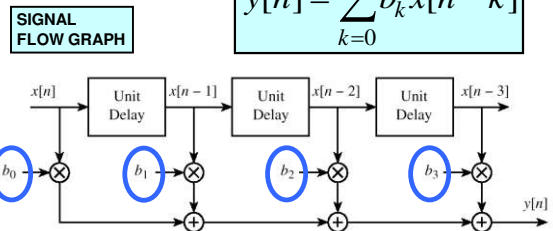
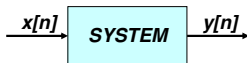


Figure 5.13 Block-diagram structure for the M th order FIR filter.

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SYSTEM PROPERTIES



- MATHEMATICAL DESCRIPTION
- TIME-INVARIANCE
- LINEARITY
- CAUSALITY
 - “No output prior to input”

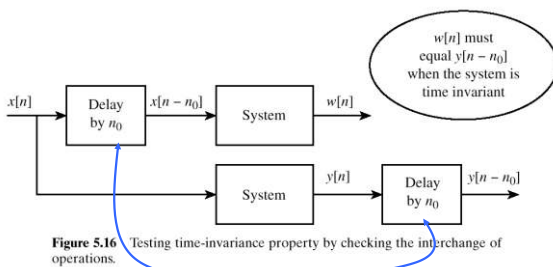
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TIME-INVARIANCE

- IDEA:
 - “Time-Shifting the input will cause the **same** time-shift in the output”
- EQUIVALENTLY,
 - We can prove that
 - The time origin ($n=0$) is picked arbitrary

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TESTING Time-Invariance



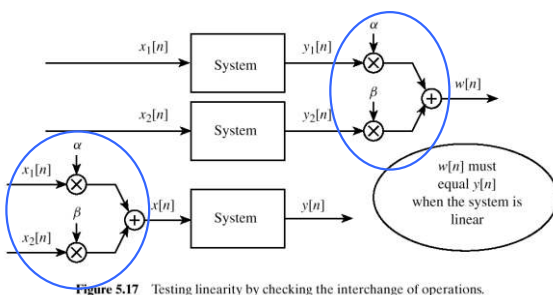
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LINEAR SYSTEM

- LINEARITY = Two Properties
- SCALING
 - “Doubling $x[n]$ will double $y[n]$ ”
- SUPERPOSITION:
 - “Adding two inputs gives an output that is the sum of the individual outputs”

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TESTING LINEARITY



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LTI SYSTEMS

- LTI: **L**inear & **T**ime-Invariant
- COMPLETELY CHARACTERIZED by:
 - IMPULSE RESPONSE $h[n]$
 - CONVOLUTION: $y[n] = x[n] * h[n]$
 - The “rule” defining the system can ALWAYS be re-written as convolution
- FIR Example: $h[n]$ is same as b_k

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POP QUIZ

- FIR Filter is “FIRST DIFFERENCE”
 - $y[n] = x[n] - x[n-1]$
- Write output as a convolution
 - Need impulse response

$$h[n] = \delta[n] - \delta[n-1]$$

- Then, another way to compute the output:

$$y[n] = (\delta[n] - \delta[n-1]) * x[n]$$

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CASCADE SYSTEMS

- Does the order of S_1 & S_2 matter?
 - NO, LTI SYSTEMS can be rearranged !!!
 - WHAT ARE THE FILTER COEFFS? $\{b_k\}$

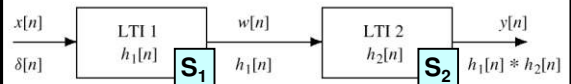


Figure 5.19 A Cascade of Two LTI Systems.

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CASCADE EQUIVALENT

- Find “overall” $h[n]$ for a cascade ?

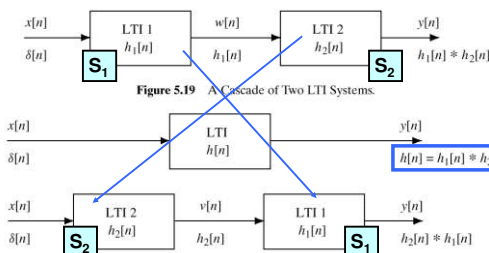


Figure 5.20 Switching the order of cascaded LTI systems.

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DOMAINS: Time & Frequency

- Time-Domain: “ n ” = time
 - $x[n]$ discrete-time signal
 - $x(t)$ continuous-time signal
- Frequency Domain (sum of sinusoids)
 - Spectrum vs. f (Hz)
 - ANALOG vs. DIGITAL
 - Spectrum vs. ω (rad/s)
- Move back and forth QUICKLY

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FREQUENCY RESPONSE

- INPUT: $x[n]$ = SINUSOID
- OUTPUT: $y[n]$ will also be a SINUSOID
 - Different Amplitude and Phase
 - SAME Frequency
- AMPLITUDE & PHASE CHANGE
 - Called the FREQUENCY RESPONSE

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COMPLEX EXPONENTIAL

$$x[n] = Ae^{j\phi} e^{j\hat{\omega}n} \quad -\infty < n < \infty$$

$x[n]$ is the input signal—a complex exponential

$$y[n] = \sum_{k=0}^M b_k x[n-k] = \sum_{k=0}^M h[k] x[n-k]$$

FIR DIFFERENCE EQUATION

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COMPLEX EXP OUTPUT

- Use the FIR “Difference Equation”

$$\begin{aligned}
 y[n] &= \sum_{k=0}^M b_k x[n-k] = \sum_{k=0}^M b_k A e^{j\varphi} e^{j\hat{\omega}(n-k)} \\
 &= \left(\sum_{k=0}^M b_k e^{j\hat{\omega}(-k)} \right) A e^{j\varphi} e^{j\hat{\omega}n} \\
 &= H(\hat{\omega}) A e^{j\varphi} e^{j\hat{\omega}n}
 \end{aligned}$$

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FREQUENCY RESPONSE

- At each frequency, we can **DEFINE**

$$H(e^{j\hat{\omega}}) = \sum_{k=0}^M b_k e^{-j\hat{\omega}k} \quad \leftarrow \text{FREQUENCY RESPONSE}$$

- Complex-valued formula
 - Has **MAGNITUDE** vs. frequency
 - And **PHASE** vs. frequency
- Notation: $H(e^{j\hat{\omega}})$ in place of $H(\hat{\omega})$

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EXAMPLE 6.1

$$\{b_k\} = \{1, 2, 1\}$$

$$\begin{aligned}
 H(e^{j\hat{\omega}}) &= 1 + 2e^{-j\hat{\omega}} + e^{-j2\hat{\omega}} \\
 &= e^{-j\hat{\omega}} (e^{j\hat{\omega}} + 2 + e^{-j\hat{\omega}}) \\
 &= e^{-j\hat{\omega}} (2 + 2\cos\hat{\omega})
 \end{aligned}$$

EXPLOIT SYMMETRY

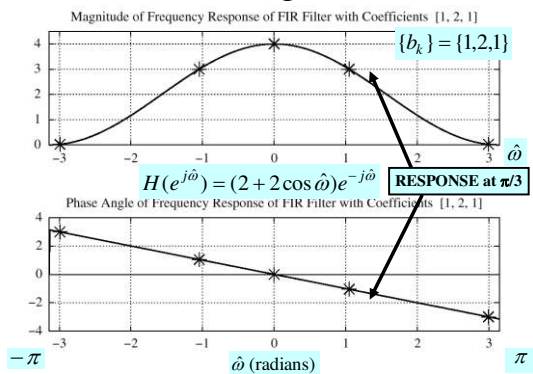
Since $(2 + 2\cos\hat{\omega}) \geq 0$

Magnitude is $|H(e^{j\hat{\omega}})| = (2 + 2\cos\hat{\omega})$

and Phase is $\angle H(e^{j\hat{\omega}}) = -\hat{\omega}$

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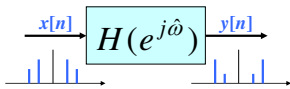
PLOT of FREQ RESPONSE



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EXAMPLE 6.2

Find $y[n]$ when $H(e^{j\hat{\omega}})$ is known
and $x[n] = 2e^{j\pi/4} e^{j(\pi/3)n}$



$$H(e^{j\hat{\omega}}) = (2 + 2\cos\hat{\omega})e^{-j\hat{\omega}}$$

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EXAMPLE 6.2 (answer)

Find $y[n]$ when $x[n] = 2e^{j\pi/4} e^{j(\pi/3)n}$

One Step - evaluate $H(e^{j\hat{\omega}})$ at $\hat{\omega} = \pi/3$

$$H(e^{j\hat{\omega}}) = (2 + 2\cos\hat{\omega})e^{-j\hat{\omega}}$$

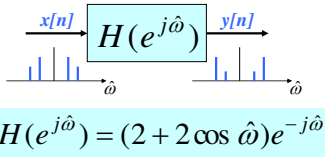
$$H(e^{j\hat{\omega}}) = 3e^{-j\pi/3} \quad @ \hat{\omega} = \pi/3$$

$$y[n] = (3e^{-j\pi/3}) \times 2e^{j\pi/4} e^{j(\pi/3)n} = 6e^{-j\pi/12} e^{j(\pi/3)n}$$

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EXAMPLE: COSINE INPUT

Find $y[n]$ when $H(e^{j\hat{\omega}})$ is known and $x[n] = 2 \cos(\frac{\pi}{3}n + \frac{\pi}{4})$



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EX: COSINE INPUT

Find $y[n]$ when $x[n] = 2 \cos(\frac{\pi}{3}n + \frac{\pi}{4})$

$$2 \cos(\frac{\pi}{3}n + \frac{\pi}{4}) = e^{j(\pi n/3 + \pi/4)} + e^{-j(\pi n/3 + \pi/4)}$$

$$\Rightarrow x[n] = x_1[n] + x_2[n]$$

Use
Linearity

$$y_1[n] = H(e^{j\pi/3})e^{j(\pi n/3 + \pi/4)}$$

$$y_2[n] = H(e^{-j\pi/3})e^{-j(\pi n/3 + \pi/4)}$$

$$\Rightarrow y[n] = y_1[n] + y_2[n]$$

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EX: COSINE INPUT (ans-2)

Find $y[n]$ when $x[n] = 2 \cos(\frac{\pi}{3}n + \frac{\pi}{4})$

$$H(e^{j\hat{\omega}}) = (2 + 2 \cos \hat{\omega})e^{-j\hat{\omega}}$$

$$y_1[n] = H(e^{j\pi/3})e^{j(\pi n/3 + \pi/4)} = 3e^{-j(\pi/3)}e^{j(\pi n/3 + \pi/4)}$$

$$y_2[n] = H(e^{-j\pi/3})e^{-j(\pi n/3 + \pi/4)} = 3e^{j(\pi/3)}e^{-j(\pi n/3 + \pi/4)}$$

$$y[n] = 3e^{j(\pi n/3 - \pi/12)} + 3e^{-j(\pi n/3 - \pi/12)}$$

$$\Rightarrow y[n] = 6 \cos(\frac{\pi}{3}n - \frac{\pi}{12})$$

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MATLAB: FREQUENCY RESPONSE

• **HH = freqz(bb, 1, ww)**

– VECTOR **bb** contains Filter Coefficients

• <https://www.mathworks.com/help/signal/ref/freqz.html>

• FILTER COEFFICIENTS $\{b_k\}$

$$H(e^{j\hat{\omega}}) = \sum_{k=0}^M b_k e^{-j\hat{\omega}k}$$

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Time & Frequency Relation

• Get Frequency Response from $h[n]$

– Here is the FIR case:

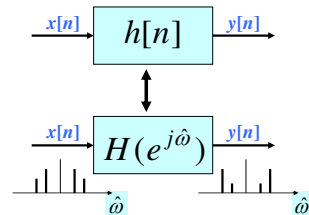
$$H(e^{j\hat{\omega}}) = \sum_{k=0}^M b_k e^{-j\hat{\omega}k} = \sum_{k=0}^M h[k] e^{-j\hat{\omega}k}$$

IMPULSE RESPONSE

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BLOCK DIAGRAMS

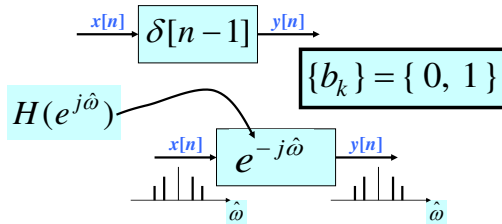
• Equivalent Representations



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UNIT-DELAY SYSTEM

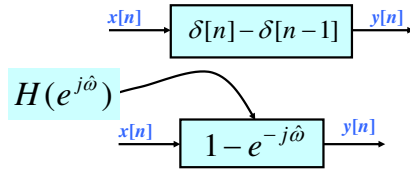
Find $h[n]$ and $H(e^{j\hat{\omega}})$ for $y[n] = x[n - 1]$



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FIRST DIFFERENCE SYSTEM

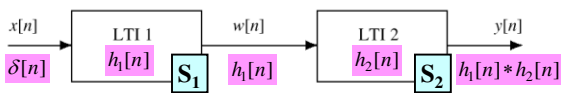
Find $h[n]$ and $H(e^{j\hat{\omega}})$ for the Difference Equation: $y[n] = x[n] - x[n - 1]$



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CASCADE SYSTEMS

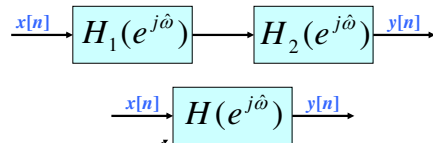
- Does the order of S_1 & S_2 matter?
 - NO, LTI SYSTEMS can be rearranged !!!
 - WHAT ARE THE FILTER COEFFS? $\{b_k\}$
 - WHAT is the overall FREQUENCY RESPONSE ?



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CASCADE EQUIVALENT

- MULTIPLY** the Frequency Responses



EQUIVALENT SYSTEM

$$H(e^{j\hat{\omega}}) = H_1(e^{j\hat{\omega}})H_2(e^{j\hat{\omega}})$$

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